HOW IT WORKS

How a print job runs a language model

TONER-1 is the most capable language model ever to run entirely inside a print job.* It is one 595 KB PostScript file. Send it to a laser printer and the printer's own processor writes a short story, usually about a printer, then prints it. The computer that sent it does no math.

01

A print job is a program

PostScript, the language most laser printers speak, is not a file format but a complete programming language, stack-based like Forth. Send `3 4 add` and the interpreter leaves 7 on its stack. It has variables, loops and floating-point math. A print job is simply a program; the printed page is a side effect of the drawing commands it calls.

02

Printers are computers

To run those programs, every PostScript printer carries its own processor and memory. The 1985 Apple LaserWriter had a 12 MHz Motorola 68000 and 1.5 MB of RAM, a faster CPU than the original Macintosh it plugged into (a 68000 at about 7.8 MHz). Modern printers have far more, and spend most of it waiting for spreadsheets.

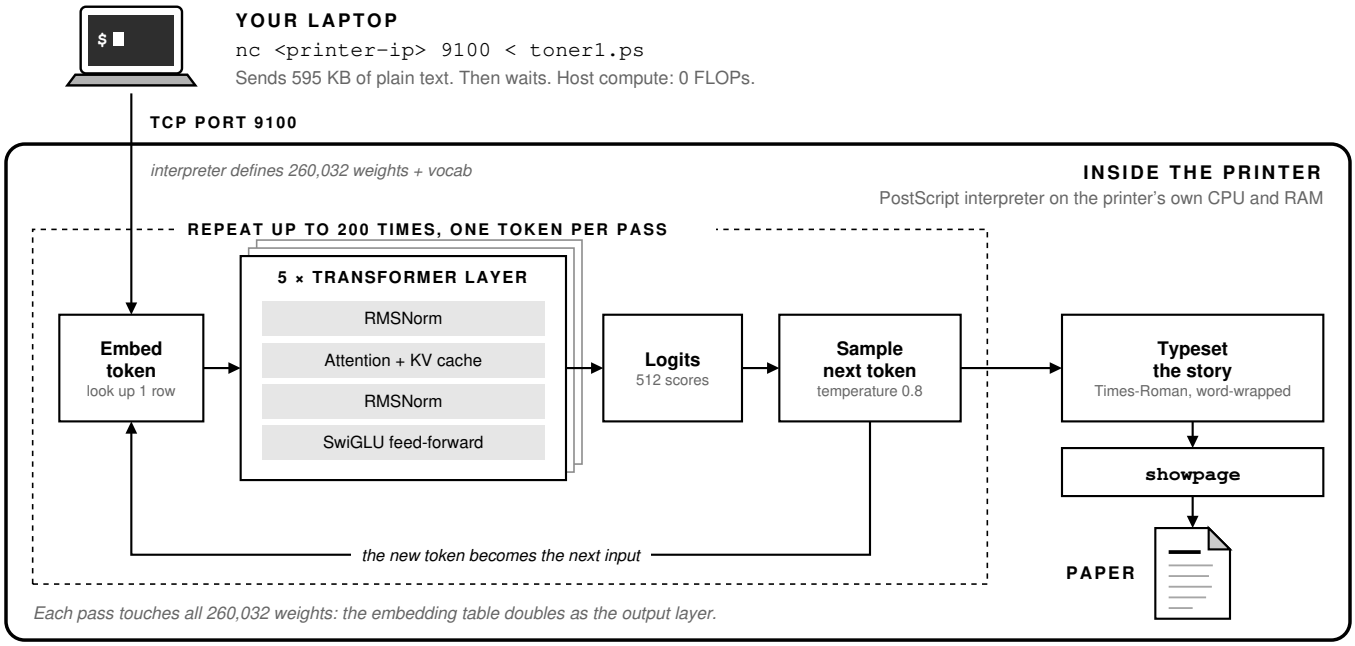
03

An LLM is just arithmetic

Strip away the mystique and a language model is a long list of multiplications and additions: matrix-vector products, a normalization step, a softmax that turns scores into probabilities, and a weighted coin flip that picks the next word piece. Any language with loops and multiplication can run one. PostScript has both.

THE PIPELINE

What happens after you send toner1.ps to port 9100



260,032
parameters
int8, one scale per row

595 KB
the entire print job
weights + vocab + program

≈80 tok/s
in Ghostscript 10.07
Apple-silicon Mac, measured

0.92 tok/s
on a real printer
HP M281fdw, measured

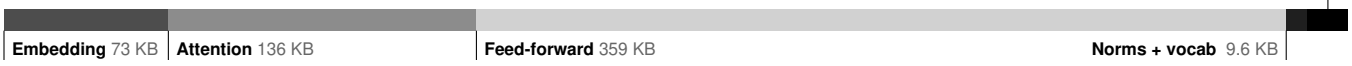
0 FLOPs
on the host
the computer only sends bytes

** Among int8, 260,032-parameter, English-language, children's-story models executed by a PostScript interpreter, as of this printing. Ghostscript counts as a printer for the purposes of this footnote. Model: Karpathy's llama2.c stories260K, trained on TinyStories, then fine-tuned by us on 8,757 printer stories written by Xing4.0-29B-A4B (Apache-2.0). Not affiliated with or endorsed by Adobe, HP, Artifex, Google, or any AI lab.*

INSIDE THE FILE

What's inside the 595 KB print job

BYTES IN TONER1.PS BY PURPOSE: 595,372 TOTAL, 455 LINES OF PLAIN TEXT



THE DATA, LINES 1 TO 106

```
 1 %!PS-Adobe-3.0
 7 /D 64 def /HID 172 def /NL 5 def /NH 8 def /NKV 4 def
     /V 512 def /SEQ 512 def /HS 8 def /KVD 32 def
     /KVMUL 2 def /NPARAMS 260032 def
```

```
16 /WQ [
17 [[0.00261005 0.00431596 0.00340043 0.00252913 ... 0.0012534]
18 [<96917e84906c8a8537147183bfa939875d25794a...> <ae99...> ...]]
```

```
row 0 bytes        96       91       7e       84
minus 128          22       17       -2        4
× 0.00261005   0.0574   0.0444  -0.0052   0.0104
```

1. **The model card is one line.** Line 7 defines width 64, 5 layers, 8 query heads sharing 4 key/value heads and a 512-token vocabulary. Nothing runs yet; names are just defined. Line 1 makes it a print job; lines 2 to 6 are comments for the spooler.

2. **Weights are bytes.** For each matrix row, export.py takes the largest absolute weight m and stores scale = m/127. Each weight is rounded to an integer from -127 to 127, plus 128, and written as one byte of a hex string. Two hex characters per weight: that is most of the file.

THE PROGRAM, LINES 107 TO 455

```
129 % o x M mv  ;  M = [scales rows], rows ... offset by 128
130 /mv { /mM exch def /mx exch def /mo exch def
131   /mS mM 0 get def /mQ mM 1 get def /mn mx length 1 sub def
132   /msx 0 mx { add } forall 128 mul def
133   0 1 mQ length 1 sub { /mi exch def /mrow mQ mi get def
134     0 0 1 mn { dup mrow exch get exch mx exch get mul add } for
135     msx sub mS mi get mul mo mi 3 -1 roll put } for
136 } bind def
```

3. **mv does the heavy lifting.** Matrix times vector: about 259,000 multiply-adds per token run through these eight lines. `get` on a string returns a byte as an integer. Rather than subtract 128 from every byte, mv takes the raw dot product, subtracts `msx` (128 times the sum of the input), then applies the row's scale.

```
139 /rope { /rn exch def /rv exch def
140   0 2 rn 1 sub { /ri exch def
141     /hd ri HS mod def
142     pos 10000 hd HS div exp div 57.29577951 mul /ang exch def
143     /fcr ang cos def /fci ang sin def
```

4. **RoPE, in degrees.** Rotary embeddings turn query/key pairs by `pos / 10000^(hd/8)` radians, but PostScript's `sin` and `cos` take degrees. Hence 57.29577951 (180/pi). Without it, every position would get the wrong rotation.

```
118 SEED srand 4 { rand pop } repeat
204 /r rand 2147483647 div ssum mul def /cum 0 def
      /pick V 1 sub def
205 0 1 V 1 sub { /j exch def /cum cum logits j get add def
      cum r gt { /pick j def exit } if } for
```

5. **Sampling with rand.** The logits become a softmax at temperature 0.8. PostScript's built-in `rand` picks a point along the total, and the loop walks the running sum until it passes it. Line 118 skips four tiny first draws. Seed 42: same interpreter, same story.

```
257 % ... typesetting: the same print job lays out the page ...
368 fs /Times-Roman F ...
449 9 /Helvetica-Bold F 72 79 moveto (Host compute: ...) show
453 showpage
```

6. **Then, and only then, toner.** After 200 tokens (or an end-of-story token), the same job word-wraps the story in Times-Roman, draws the run card and calls `showpage`. Until then, the printer was just a computer.

PRINT IT YOURSELF — BRING YOUR OWN PRINTER

```
nc <printer-ip> 9100 < toner1.ps
```

TONER-1 Pro: toner1_pro.ps, 12.2M parameters, 24.9 MB, ours. Same command. About 28 times slower.

- **Needs real PostScript.** A PostScript interpreter, Level 2 or later, is required. PCL- or GDI-only printers may print the file as raw text.
- **Mind the limits.** Long jobs can hit the job timeout or the VM limit. In Ghostscript, TONER-1 uses 3.3 to 7.5 MB of VM.
- **Expect minutes.** An HP M281fdw took 3:37 for 200 tokens by its own clock, 0.92 tok/s. It is thinking.
- **Ask first.** Only use printers you own or have permission to use. The queue behind you has feelings.

STATUS ☑ Verified in Ghostscript 10.07: 200 tokens, about 80 tok/s ☑ **Printed on an HP M281fdw, 2026-11-05: 200 tokens in 3:37**

Excerpts are verbatim from toner1.ps; gray . . . marks trims, and long lines are re-wrapped or dedented. Prior art: int-llm-postscript (Aug 2026) ran a 14K-parameter microGPT in Ghostscript only; PSChess played chess on real printers. On 2026-11-05 TONER-1 ran on an actual printer, an HP Color LaserJet Pro MFP M281fdw. Not affiliated with or endorsed by Adobe, HP, Artifex, Google, or any AI lab.